Classification Assignment – Chronic Kidney Disease Prediction

1. Problem Statement

To predict whether a patient has Chronic Kidney Disease (CKD) based on medical information

2. Solution

I used classification models and compared their performance using Accuracy, Precision, Recall, and F1 Score

3. Data Preprocessing

- Checked for missing values and duplicates.
- Converted categorical columns into numerical values using One-Hot Encoding.
- Applied StandardScaler to numerical features.
- Split the dataset

4. Algorithms I Used

- Logistic Regression
- Decision Tree Classification
- Random Forest Classification
- Gradient Boosting Classification
- Support Vector Machine

5. Model Comparison

Algorithm	Accuracy
Logistic Regression	98.75%
Decision Tree	98.75%
Random Forest	98.75%
Gradient Boosting	98.75%
SVM	98.75%

6. Logistic Regression

I selected Logistic Regression because all five algorithms achieved the same accuracy of 98.75%.